



A generalizable automated geophysical agent workflow for accessible subsurface hydrology analysis

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ABSTRACT

Geophysical methods are central to subsurface imaging, yet their end-to-end workflows from data ingestion and inversion to petrophysical conversion remain inaccessible to non-specialists due to steep software and methodological learning curves. Large Language Models (LLMs) are increasingly used as code assistants in scientific computing, but this approach still assumes that domain experts design and maintain analytical pipelines themselves. Here, we introduce a Generalizable Automated Geophysical Agent Workflow (GAGAW) that reframes LLMs from code generators to domain-aware, multi-agent workflow orchestrators. GAGAW translates plain-language scientific intents into structured configurations that coordinate specialized open-source tools for geophysical data processes and interpretation. We demonstrate GAGAW in hydrogeophysics through three widely adopted workflows: petrophysical conversion, time-lapse inversion, and multi-source data fusion. In these examples, brief natural language prompts initiated complete analytical pipelines that would traditionally require extensive scripting expertise, producing reproducible resistivity models, water-content estimates, climate analyses, and comprehensive reports. Based on this workflow, we developed Autonomous Query-driven Understanding Agent for Hydrogeophysics (AQUAH), a web-based platform providing an intuitive interface for non-programmers. I also outline current limitations and future directions for geophysical agents. GAGAW establishes a practical pattern for accessible, cross-modal geophysical analysis and marks a paradigm shift from LLM-assisted scripting to LLM-orchestrated, tool-centric workflows in Earth and environmental science.

1. Introduction

Geophysical methods have a long and successful history as powerful tools for understanding the subsurface (Kearey et al., 2002). They provide minimally invasive and spatially extensive techniques to image subsurface structures and properties. These approaches span gravity, magnetic, seismic, and electromagnetic sensing across decimeter-to-kilometer scales, enabling 1D, 2D, 3D, and increasingly 4D (time-lapse) imaging (e.g., Chen et al., 2024; Finn et al., 2022; Izzeldin, 1987). These methods play an important role in a wide range of applications, including mineral exploration, groundwater assessment, geothermal and hydrogen energy exploration, oil and gas reservoir monitoring, natural hazard assessment, archaeological investigations, and environmental remediation (Binley et al., 2015; Mao et al., 2025; Oldenburg et al., 1998; Parsekian et al., 2015; Wright et al., 1985). The ability to infer subsurface properties from surface-based or borehole measurements makes these techniques indispensable for understanding Earth processes.

However, the geophysical data processing workflow presents a steep learning curve for non-specialists. It involves managing diverse data formats, signal processing, forward modeling, and inversion to con-

vert measurements into subsurface physical models, and each step requires a deep understanding of geophysics. A growing ecosystem of open-source software aims to lower the barrier to entry for geophysical analysis; for instance, the Python landscape offers versatile inversion libraries like SimPEG and pyGIMLi (Cockett et al., 2015; Rücker et al., 2017), specialized packages for electrical data processing such as ResIPy (Blanchy et al., 2020), frameworks for 3D geological modeling like GemPy (De la Varga et al., 2019), and software accelerates seismic wave modeling like Pyseis (Farris et al., 2023). However, users still need to navigate a steep learning curve for each individual tool. Because different packages offer complementary features, a comprehensive investigation often requires chaining multiple programs together, a task complicated by their distinct programming interfaces and data conventions, creating inconveniences even for specialists.

The recent emergence of powerful Large Language Models (LLMs) such as ChatGPT has introduced a new paradigm for human-computer interaction and scientific inquiry (Achiam et al., 2023). LLMs have been applied across various scientific domains, including drug discovery and clinical applications in medical sciences (Lu et al., 2025; Thirunavukarasu et al., 2023), and the engineering area in electronics

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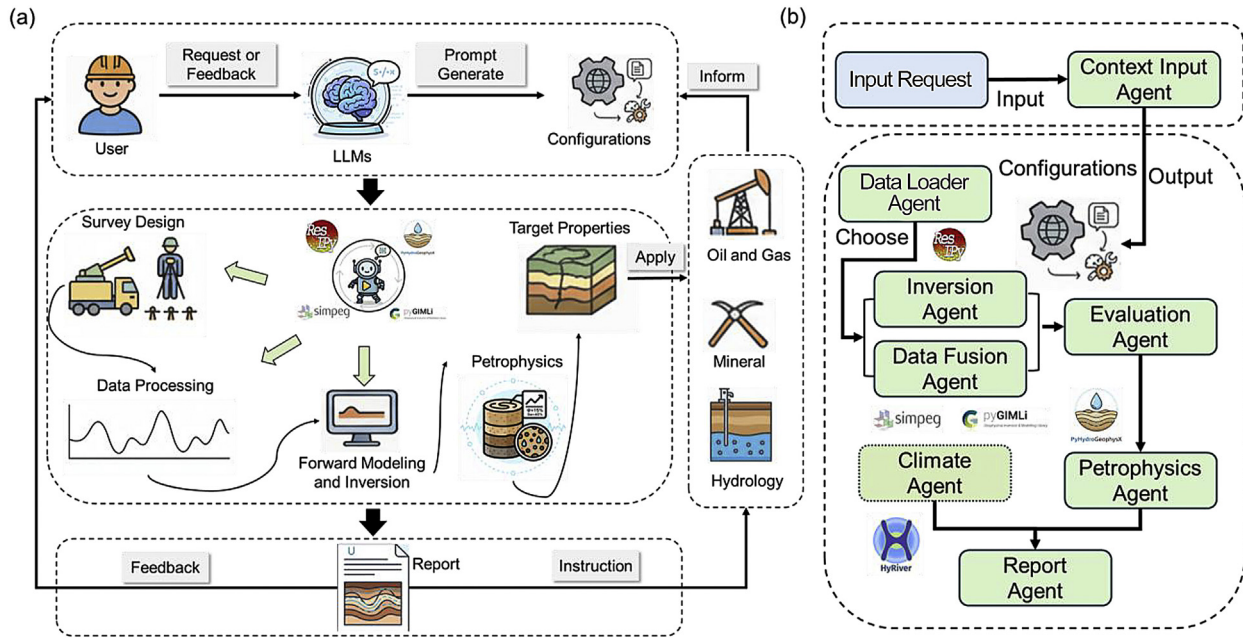


Fig. 1. The Generalizable Automated Geophysical Agent Workflow (GAGAW). (a) The complete workflow begins with a user's natural language request, which an LLM translates into a configuration file. This file directs the GAGAW agent to perform a series of tasks, including survey design, data processing, inversion, and petrophysical analysis with a suite of open-source Python packages. The final output is a report that can be iteratively refined based on user feedback. (b) The specific multi-agent system is designed for the hydrogeophysical application in this study. It details the roles of individual agents, such as the Data Loader Agent, the Inversion Agent, the Petrophysics Agent, and the optional Climate Agent, as they work together to convert geophysical data into hydrologic properties.

and chip design (Alsaqer et al., 2024; Chang et al., 2024). LLMs in scientific computing were primarily used as code-generation assistants, helping researchers write individual scripts or debug syntax errors (Wang & Chen, 2023). However, this code-centric approach still requires that domain experts possess deep knowledge of the underlying tools and manually design the analytical pipelines. More recently, these LLMs have been integrated into autonomous AI agents: systems that can observe their environment, make decisions, and take actions to achieve specific goals (Li et al., 2024). Agent is capable of managing complete end-to-end workflows as orchestrators rather than mere assistants. Prompt engineering can manage these agents, allowing users to effectively guide the agents' behavior by designing specific input prompts that optimize their performance in complex tasks (Chen B. et al., 2023). These agents are now being applied across disciplines to automate complex workflows, from multimodal 3D brain CT report generation in healthcare to predicting crystalline materials properties in materials science (Li et al., 2025; Niyongabo Rubungo et al., 2025). In these domains, agentic systems have demonstrated the ability to manage end-to-end pipelines that span data retrieval, parameter initialization, model execution, and report generation. Establishing a well-designed agent workflow significantly lowers the learning curve while maintaining accuracy, as demonstrated by its value in adjacent disciplines, such as agentic workflows lowering the technical barrier for establishing hydrologic models (Yan et al., 2025).

This work proposes a Generalizable Automated Geophysical Agent Workflow (GAGAW) that orchestrates various open-source Python frameworks to provide subsurface information (Fig. 1a). We illustrate this workflow in hydrogeophysics, a discipline that has become a critical tool for hydrologists and environmental scientists seeking to characterize the subsurface without extensive drilling (Binley et al., 2015). These methods have broad applications, including using seismic refraction to map subsurface structure (Flinchum et al., 2018) and Electrical Resistivity Tomography (ERT) for groundwater monitoring (Singha et al., 2015). For example, these techniques are used for critical zone characterization, identifying preferential flow paths in fractured rock, monitoring saltwater intrusion, and characterizing contaminant plumes (Chen H. et al., 2023; Clair et al., 2015; Kang et al., 2018; Ogilvy et al.,

2009). The process of translating raw geophysical data into these hydrogeologic insights requires the integration of diverse geophysical data, petrophysical models, site-specific calibration, and ultimately, the estimation of hydrologic properties with quantified uncertainty (Rubin & Hubbard, 2006). Each step, from instrument-specific data formatting and quality control through forward modeling, inversion regularization, and petrophysical conversion, demands specialized expertise. This entire process presents a significant learning curve for most traditional hydrologists.

Our proposed LLM agent-based framework addresses these challenges by reframing the role of LLMs in geophysics: from code-generation assistants to domain-aware workflow orchestrators. The framework automates data processing, inversion, and interpretation by integrating open-source hydrogeophysical Python packages through typed configurations and multi-agent coordination. We demonstrate this using ERT and complementary seismic refraction data to provide subsurface hydrologic information, incorporating common geophysical processes including time-lapse inversion, multi-source data fusion, and petrophysical conversions. The LLM agent manages workflows across multiple open-source geophysical and hydrological platforms, enabling comprehensive subsurface hydrology investigations without requiring users to master each individual package or maintain complex integration code. By providing a natural language interface to powerful open-source tools, the framework lowers the barrier to entry and makes cross-modal geophysical analysis practical for hydrologists and other domain experts without years of specialized training. This work represents a paradigm shift in how LLMs can serve geoscience: not as passive coding assistants, but as active analytical orchestrators that preserve physics-based rigor and human oversight while expanding methodological accessibility.

2. Methods and experiments

2.1. GAGAW framework

The GAGAW, illustrated in Fig. 1a, is an LLM-driven framework designed to translate high-level user requests into comprehensive geophysical analyses. The process initiates when a user provides a task in natural

language; using prompt engineering techniques, the LLMs, such as ChatGPT, Gemini, or DeepSeek API (Comanici et al., 2025, Guo et al., 2025; Hurst et al., 2024), interpret this input to generate a precise workflow configuration with suitable parameters for each required open-source tool, which in turn guides the subsequent steps of the analysis. This configuration directs the central GAGAW with proper prompt engineering, which could then employ a suite of open-source Python packages (e.g., pyGIMLi, SimPEG, ResIPy) to execute the required geophysical data processing and interpretation tasks. By integrating the diverse features from different open-source packages, the agent can autonomously handle the end-to-end workflow. This comprehensive process extends from initial survey design and data processing to forward modeling, inversion, and petrophysical conversion, ultimately producing target subsurface properties applicable to domains like hydrology, mineral exploration, or oil and gas. The final output is a synthesized report that the user can evaluate, providing an opportunity for feedback to iteratively refine the analysis and ensure the results align with their scientific goals.

2.2. GAGAW for hydrogeophysics

For this work, we apply the GAGAW framework to a common hydrogeophysical challenge: converting ERT data and other complementary data into meaningful hydrologic properties. Fig. 1b outlines the main agents in the workflow; additional prompt engineering details are provided in Text S1, with configuration specifics in Text S2. The workflow is executed via the ChatGPT API using GPT-4o-mini (Hurst et al., 2024). The selection of GPT-4o-mini was driven by its cost-effectiveness, with a price of just \$0.15 for input and \$0.60 for output per million tokens, which makes controlling the numerous agent interactions economically

practical. To increase the robustness of the workflow, a fallback mechanism using regular expressions is implemented to parse key information (see Text S3), in case the LLM fails under specific conditions (Kang et al., 2024).

In this hydrogeophysical application, a user request is first parsed by the Context Input Agent into a normalized workflow configuration with strict JSON fields. This configuration is produced and governed by a hierarchy of prompt-engineered templates (Figures S1 to S3; Text S1). Then, this configuration (Figures S4 to S6) drives the Data Loader Agent to select the appropriate reader, load instrument-specific ERT data, filter, and quality-check measurements for downstream steps. Depending on the configuration, results flow either to the Inversion Agent for standard or time-lapse ERT inversions or to the Data Fusion Agent. The Data Fusion Agent composes an execution plan that chains multiple agents to integrate diverse geophysical datasets using joint, coupled, or structure-constrained inversion (Linde & Doetsch, 2016) with libraries such as pyGIMLi, SimPEG, and PyHydroGeophysX (Chen et al., 2025).

The Evaluation Agent scores inversion quality; in evaluation-only mode, it evaluates and recommends, whereas in self-tuning mode, it performs a bounded set of re-inversions by adjusting parameters, retains the best result, and logs the evaluation history for transparency. For data fit, the agent targets a chi-squared value (χ^2) of approximately 1.0; values below this range indicate underfitting, while values above indicate overfitting. This chi-squared targeting ensures that data misfit matches the estimated error level, preventing the model from fitting noise. Next, the Petrophysics Agent converts resistivity to hydrologic properties using Monte Carlo sampling, returning summary statistics and an auto interpretation. The Petrophysics Agent includes built-in safeguards against physically unreasonable parameters. When the user provides petrophys-

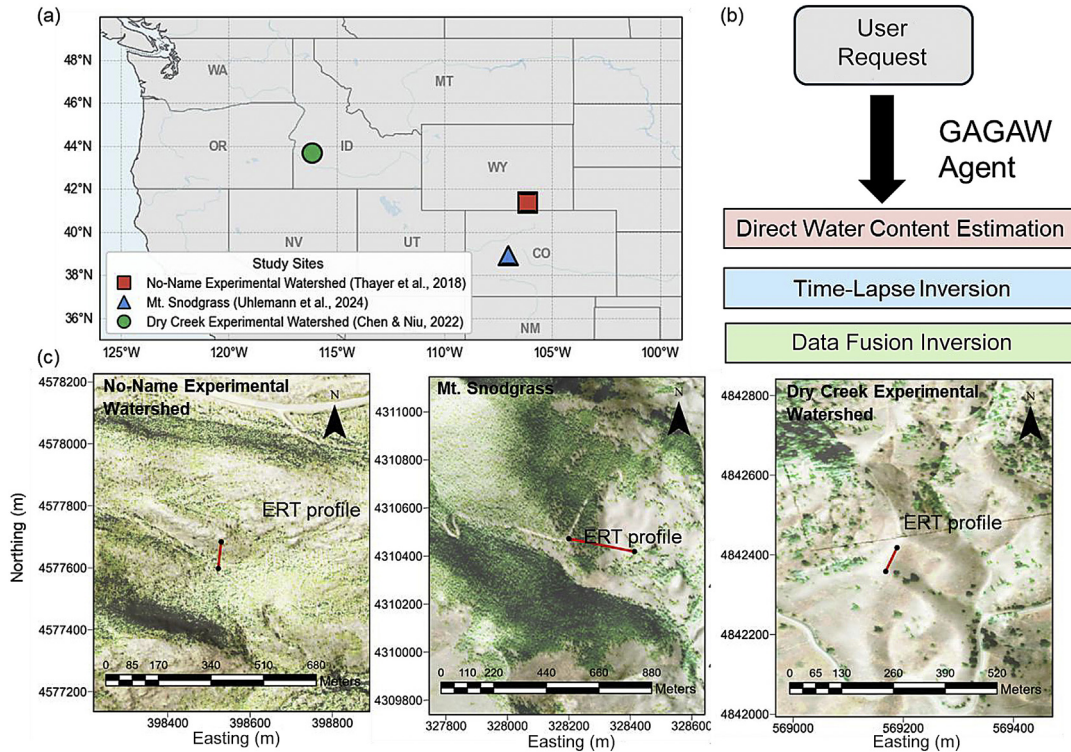


Fig. 2. Study sites and GAGAW workflow demonstrations. (a) Geographic locations of three field sites across the western United States: No-Name Experimental Watershed in Wyoming (red square), Mt. Snodgrass in Colorado (blue triangle), and Dry Creek Experimental Watershed in Idaho (green circle). (b) Three hydrogeophysical workflows orchestrated by the GAGAW agent: direct water content estimation from ERT data, time-lapse inversion coupled with meteorological forcing, and structure-constrained data fusion integrating seismic refraction with ERT. (c) High-resolution satellite imagery of each study site showing landscape characteristics and ERT profile locations (red lines with electrodes marked). Left: No-Name Experimental Watershed. Center: Mt. Snodgrass. Right: Dry Creek Experimental Watershed. Scale bars indicate spatial extent at each site.

ical values, it uses them and documents the values. When no site-specific information is provided, the agent applies default values defined in the code. When LLM-inferred parameters are used, they are constrained by physical rationality. An optional Climate Agent, particularly useful for time-lapse data analysis, can employ tools like HyRiver (Chegini et al., 2021) to gather relevant climate data such as precipitation, temperature, and evapotranspiration (ET) to add valuable context. Finally, the Report Agent synthesizes all outputs to generate a comprehensive and informative report for the user.

2.3. Experiment datasets

We tested the GAGAW workflow in subsurface hydrology applications using three field datasets spanning diverse hydrogeologic settings across the western United States (Fig. 2a). Each dataset targets one of three widely applicable hydrogeophysical workflows: direct water content estimation, time-lapse inversion, and multi-source data fusion (Fig. 2b). The first dataset comes from a snow-dominated sub-alpine hillslope in the No-Name Experimental Watershed in southeastern Wyoming (Fig. 2c, left; Thayer et al., 2018). The site is characterized by seasonal snowmelt-driven hydrology and relatively homogeneous glacial till overlying bedrock. Raw ERT data were collected in 2015 using a DAS-1 resistivity meter and are publicly available from Pleasants et al. (2022). The ERT data were collected using a DAS-1 resistivity meter, which is widely adopted for field deployments. A key challenge with this dataset, which is also a common issue in other field datasets, is that the collected ERT data file does not include topography. This requires the agent to integrate a separate electrode locations file into the ERT files. To evaluate the agent's ability to handle varying levels of user-provided information, we designed three scenarios (S1, S2, and S3) with progressively more detailed site-specific context: minimal information (S1), geological context without parameters (S2), and full site-calibrated petrophysical parameters (S3). This progression tests how effectively GAGAW can operate under different levels of prior knowledge.

The second dataset comes from a high-elevation mountainous hillslope on Mt. Snodgrass near Crested Butte, Colorado, at approximately 3,150 m elevation (Fig. 2c, center; Uhlemann, Peruzzo et al., 2024). The site exhibits strong environmental gradients, with a steep upper slope dominated by quartz monzonite porphyry bedrock and coniferous forest transitioning to a gentler lower slope of colluvium and meadow vegetation. A time-lapse ERT array monitors the site daily, capturing subsurface moisture dynamics driven by snowmelt, precipitation, and evapotranspiration. The ERT data are open-source and available in the E4D format from Uhlemann, Dafflon et al. (2024). This dataset is well-suited to test GAGAW's capability for automated time-lapse inversion coupled with climate data integration. The agent performs different inversions on multiple time steps and employs an optional Climate Agent that retrieves meteorological forcing (precipitation, temperature, potential evapotranspiration) via PyDaymet from the HyRiver framework, enabling correlation analysis between subsurface resistivity changes and hydroclimatic drivers.

The final dataset is from a granitic hillslope within the Dry Creek Experimental Watershed (DCEW) in southwestern Idaho (Fig. 2c, right; Chen & Niu, 2022). The Treeline catchment site features weathered granite overlying fractured bedrock, with heterogeneous weathering profiles that control subsurface flow partitioning. Both electrical resistivity and seismic refraction tests have been carried out along and across a ridge of the Treeline catchment in June 2020. This dataset is used to test the workflow's ability to handle a data fusion task, specifically a structure-constrained inversion. The agent first processes seismic refraction data to extract a velocity-defined interface (regolith–fractured bedrock boundary), then uses this structural constraint to guide ERT inversion, producing a more geologically realistic resistivity model. Petrophysical conversion with layer-specific parameters subsequently trans-

lates the constrained resistivity distribution into water content estimates that make use of the seismic-derived subsurface architecture.

3. Results

3.1. Direct water content estimation

We demonstrate the workflow for direct water content estimation from the No-Name Experimental Watershed ERT dataset (Pleasants et al., 2022). The Data Loader agent effectively addresses the challenge of missing topographic data by integrating a separate electrode location file into the raw ERT measurements. To evaluate GAGAW's ability to handle user requests with varying levels of detail, we designed three scenarios with progressively richer site-specific information (Fig. 3a), each initiated by a concise natural-language prompt adapted from Thayer et al. (2018):

Scenario S1 (Minimal Information): The agent received only basic inputs: data file location, electrode coordinates, instrument type (DAS-1), and a simple request: "We have ERT data from Type at File_path and electrode file in File_path and help me invert and convert to water content" (Fig. 3a, left). Without site-specific knowledge, the Petrophysics Agent applied generic default parameters for Archie's law. The resulting mean water content was 0.056 with high spatial uncertainty (mean $\sigma = 0.070$; Fig. 3b, left). The inverted resistivity model (Fig. 4a, left) shows values ranging from ~ 100 to $\sim 5,000 \Omega\text{-m}$, but the derived water content distribution (Fig. 4a, center) is uniformly low (< 0.1) across most of the profile, with correspondingly high uncertainty throughout (Fig. 4a, right).

Scenario S2 (Geological Context): Building on S1 inputs, the agent received a geological description: "The subsurface contains organic soil, coarse angular gravel and cobbles..." and was instructed to "Guess the petrophysical parameters" (Fig. 3a, center). This contextual information enabled the Petrophysics Agent to infer geology-consistent coefficients, yielding a slightly higher mean water content of 0.064 and substantially reduced uncertainty (mean $\sigma = 0.024$; Fig. 3b, center). The resistivity inversion (Fig. 4b, left) remains similar to S1, but the petrophysical conversion now produces more spatially variable water content (Fig. 4b, center) with lower uncertainty in the upper subsurface (Fig. 4b, right).

Scenario S3 (Detailed Parameters): The agent received the same geological context as S2 plus site-calibrated petrophysical parameters derived from core samples: $m = 1.3$, $n = 2.0$, $a = 0.88$, $\phi = 0.55$ (Fig. 3a, right). With this precise information, the agent produced dramatically different results: mean water content of 0.205 (closely matching the field-derived values from Thayer et al., 2018; Fig. 3b) and minimal uncertainty (mean $\sigma = 0.012$). The resistivity model (Fig. 4c, left) is nearly identical to S1 and S2, but the water content distribution (Fig. 4c, center) now shows much higher values (0.2–0.4) that align with expected saturation levels in shallow glacial till. Uncertainty is lowest throughout the profile (Fig. 4c, right, $\sigma < 0.02$), reflecting confidence in the site-specific parameterization.

The three scenarios illustrate how GAGAW's multi-agent architecture handles information flow. The Data Loader and Inversion Agents produce consistent resistivity models across scenarios (Fig. 4a–c, left panels), as they operate independently of petrophysical assumptions. In contrast, the Petrophysics Agent adapts its conversion strategy based on prompt specificity: applying generic priors (S1), inferring geology-consistent ranges (S2), or directly using calibrated values (S3). Monte Carlo sampling propagates parameter uncertainty through Archie's law, with the ensemble spread (shaded error bars in Fig. 3b, lower panel) narrowing as prior constraints tighten. The automatically generated reports (Chen, 2025) document all parameter choices, justify default selections when information is sparse, and flag high-uncertainty zones where additional calibration would improve reliability.

These results demonstrate that providing site-specific context enables GAGAW to deliver more accurate hydrogeologic interpretations. Scenario S3's agreement with field-measured water content (Fig. 3b)

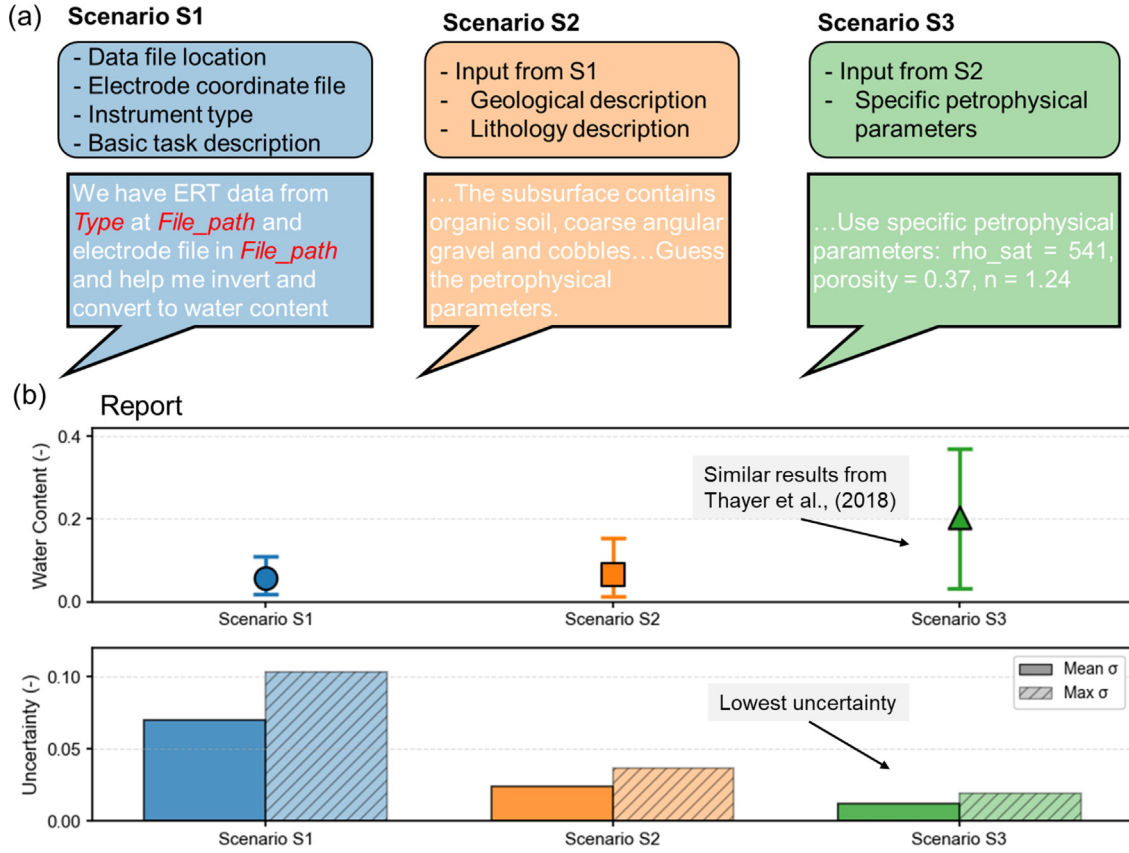


Fig. 3. Progressive refinement of water content estimation with increasing site-specific information. (a) Three scenarios with escalating input detail: S1 provides only basic data location and task description; S2 adds geological context and requests parameter inference; S3 supplies calibrated petrophysical coefficients. Natural-language prompts (lower boxes) show actual user inputs to the GAGAW system, where red italic text indicates placeholder variables (*Type*, *File_path*), replaced with the specific instrument model and data file locations. (b) Automated report outputs showing mean water content estimates and uncertainty statistics. The S3 estimate closely matches the field-derived value from [Thayer et al. \(2018\)](#). Uncertainty decreases systematically as prior information strengthens.

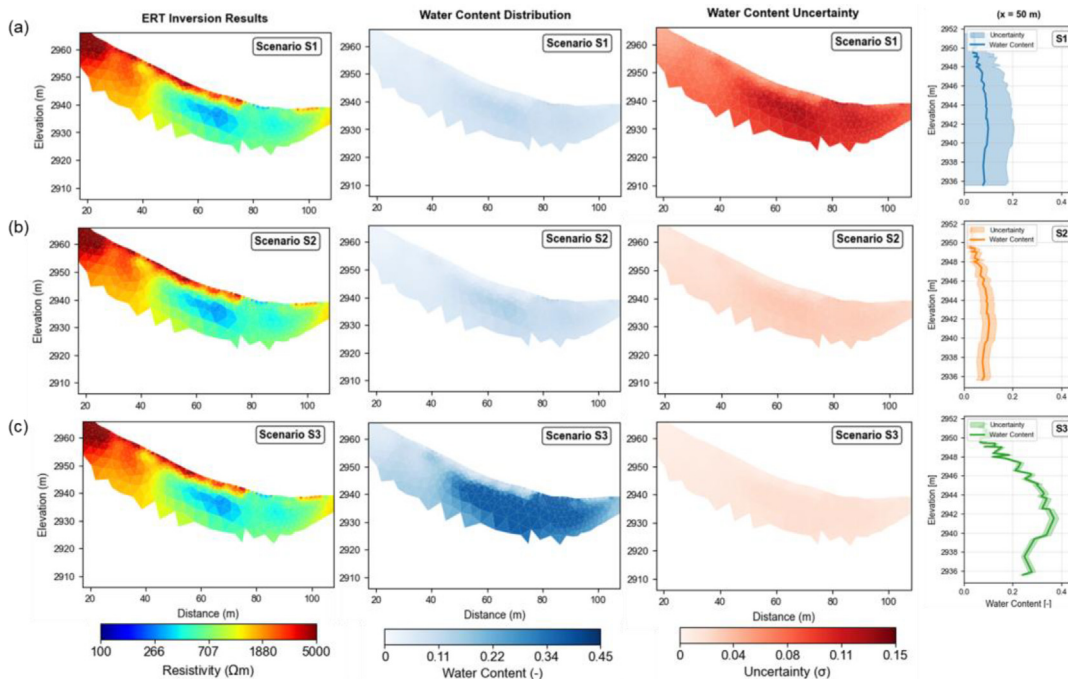


Fig. 4. Cross-sectional visualizations of inverted resistivity, water content, and uncertainty for three scenarios. (a)–(c), Results for scenarios S1, S2, and S3, respectively. First column: Electrical resistivity tomography inversion. Second column: Water content distributions. Third column: Spatial uncertainty maps (σ in water content). Fourth column: Vertical profiles at $x = 50$ m showing water content (solid line) with uncertainty bands (shaded area) for direct comparison across scenarios.

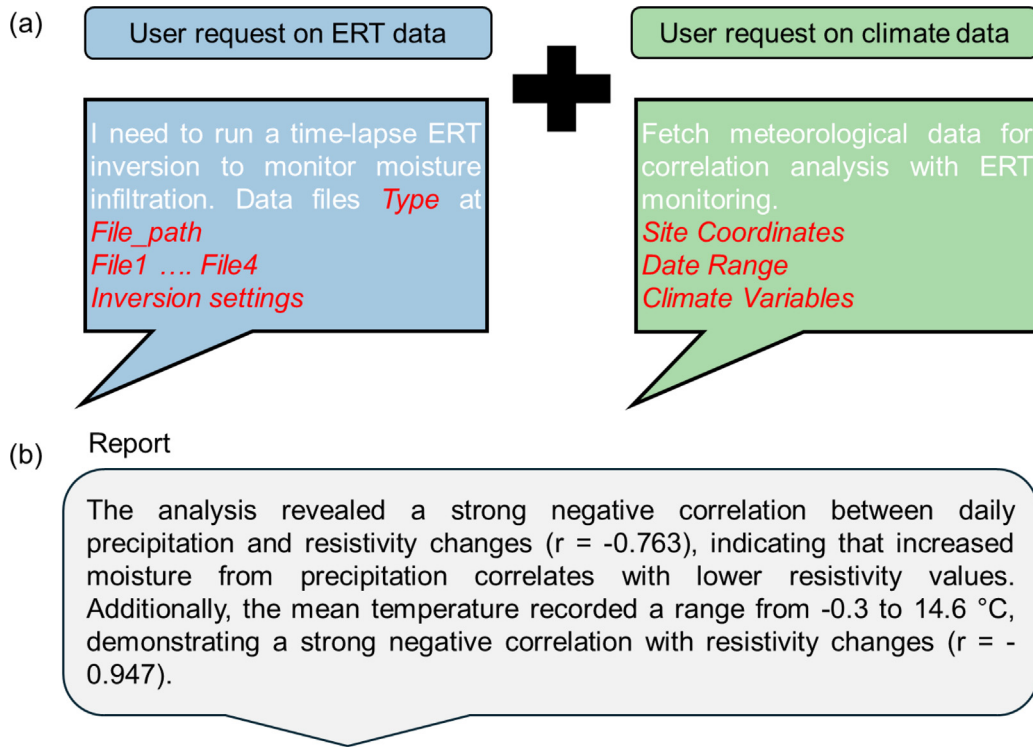


Fig. 5. Time-lapse workflow initialization and automated correlation analysis. (a) Dual natural-language prompts coupled with geophysical-climate analysis. The left (blue) box requests time-lapse ERT inversion of four monitoring surveys; the right (green) box requests meteorological data retrieval for the same period. Red italic text indicates user-configurable placeholders. The “+” symbol denotes that GAGAW processes both requests as a unified workflow. (b) Excerpt from the automatically generated report.

validates the agent’s ability to correctly apply expert-provided parameters, while S1 and S2 illustrate lower performance when operating with limited information. This pattern arises because the resistivity inversion is robust across information levels, but the downstream petrophysical conversion, the most critical step for hydrologic applications, depends strongly on domain knowledge encoded in the natural-language prompt. A synthetic benchmark test with known ground truth water content further validates the workflow’s geophysical soundness (Text S4, Figures S7–S9). Beyond accuracy, the uncertainty quantification provided by GAGAW (S1–S3) serves an important educational function for hydrologists and other non-specialists: it helps users understand how reliable geophysical results are and builds trust in geophysical tools by making the interpretation process transparent. Importantly, when uncertainty is high, this does not indicate that geophysics is unreliable; rather, it signals that insufficient site-specific information has been provided. This transparency helps users recognize that improved results require better input constraints, thereby promoting a more informed use of geophysical methods in hydrological applications.

3.2. Time-lapse data analysis

We applied GAGAW to a time-lapse ERT monitoring experiment at the Mt. Snodgrass hillslope (Uhlemann, Dafflon et al., 2024), coupling subsurface electrical imaging with daily meteorological forcing. Two concise natural-language prompts initiated the workflow (Fig. 5a): (1) a request for time-lapse inversion of four E4D-format datasets (blue box): “I need to run a time-lapse ERT inversion to monitor moisture infiltration. Data files *Type* at *File_path*, *File1 File4*, *Inversion settings*,” and (2) a parallel request for climate data integration (green box): “Fetch meteorological data for correlation analysis with ERT monitoring. *Site Coordinates*, *Date Range*, *Climate Variables*.” The red italic terms represent user-specified configurations (instrument type, file paths, coordinates, temporal bounds, and desired climate variables). The natural-language site description was adapted from Uhlemann, Peruzzo et al. (2024). The Con-

text Input Agent parsed these prompts into a structured JSON configuration (Figure S5) specifying temporal bounds (February–July 2022), instrument type (E4D), site coordinates (38.82584°N , $106.97998^{\circ}\text{W}$), and required climate covariates (precipitation, minimum/maximum temperature, potential evapotranspiration via Penman–Monteith).

The Data Loader Agent prepared four time-lapse datasets, established the baseline survey (2022-03-26), and configured difference-inversion regularization. The Inversion Agent executed time-lapse inversions, computing resistivity changes relative to the snowmelt baseline. Simultaneously, the Climate Agent invoked PyDaymet from HyRiver to retrieve daily meteorological observations across the survey period and calculated reference evapotranspiration using the Penman–Monteith formulation. The Report Agent then synthesized geophysical and climate data into synchronized visualizations and performed automated correlation analysis.

Fig. 6a shows the retrieved climate forcing aligned with ERT survey dates. Potential evapotranspiration (orange line) exhibits strong seasonal trends with daily values ranging from ~ 1 – 5 mm/day, while precipitation (blue bars) shows episodic events concentrated in spring and early summer, with several multi-day storms delivering 10–15 mm. Temperature variations (Fig. 6b) reveal a transition from late-winter conditions (minimum temperatures near -10°C in March) to summer warming (maximum temperatures exceeding 20°C by July), with survey dates (squares/circles on temperature envelopes) capturing key phases of snowmelt and infiltration. The shaded bands represent daily min/max ranges, and the correspondence between ERT surveys (marked symbols) and temperature trends enables direct interpretation of thermal versus moisture-driven resistivity changes.

Fig. 6c presents the baseline resistivity model (left panel, 2022-03-26) showing values from ~ 30 $\Omega\cdot\text{m}$ to $\sim 3,000$ $\Omega\cdot\text{m}$, with higher resistivities in the shallow subsurface consistent with partially frozen or dry soil conditions. The three subsequent time steps reveal systematic resistivity reductions (negative $\Delta\rho/\rho$ values, blue colors) concentrated in the upper 5–10 m. Time Step 1 (2022-04-26) exhibits localized decreases up to

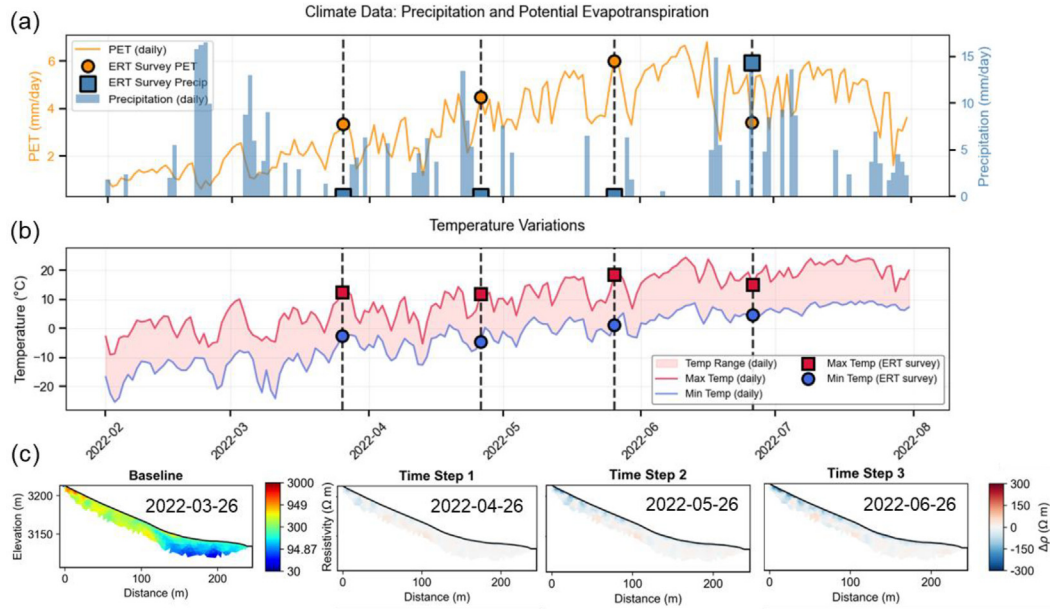


Fig. 6. Synchronized climate forcing and time-lapse resistivity response at Mt. Snodgrass. (a) Daily potential evapotranspiration and precipitation retrieved via PyDaymet, with vertical dashed lines marking ERT survey dates. (b) Daily temperature envelope showing minimum (blue) and maximum (pink) temperatures and survey-date markers (squares/circles). (c) Baseline absolute resistivity model (left, 2022-03-26) and three time-lapse difference sections showing resistivity changes ($\Delta\rho$, Ωm) at subsequent surveys.

-150 $\Delta\rho/\rho$ (Ωm) in shallow zones, coinciding with initial snowmelt and temperature increases above freezing. Time Step 2 (2022-05-26) shows intensified and spatially expanded resistivity reductions exceeding -200 $\Delta\rho/\rho$ (Ωm), consistent with sustained infiltration from snowmelt and spring precipitation events visible in Fig. 6a. Time Step 3 (2022-06-26) maintains strong negative anomalies, though spatial patterns shift toward deeper penetration, suggesting progressive wetting of the subsurface as the active layer deepens and preferential flow pathways conduct moisture downward.

The Report Agent quantified relationships between climate forcing and subsurface response (Fig. 5b). The analysis revealed a strong negative correlation between daily precipitation and resistivity change ($r = -0.763$), indicating that increased moisture from precipitation events correlates with lower resistivity values, as expected from Archie's law. Additionally, mean temperature exhibited an even stronger negative correlation with resistivity changes ($r = -0.947$), demonstrating that warming drives resistivity reductions through the combined effects of ice-to-water phase transitions and enhanced infiltration from snowmelt. The temperature range spanning -0.3°C to 14.6°C brackets the freezing point, confirming that observed resistivity dynamics reflect hydrologic rather than purely thermal processes.

From the plain-language prompts totaling ~ 100 words, GAGAW autonomously executed a multimodal workflow encompassing: (1) loading and quality-checking four time-series ERT datasets, (2) configuring and running time-lapse inversions with appropriate temporal regularization, (3) retrieving the meteorological observations and PET via a physically based model, (4) synchronizing geophysical and climate time series, (5) calculating correlation metrics with statistical significance, and (6) generating narrative reports with publication-ready figures (Chen, 2025). The entire pipeline was completed without manual intervention, demonstrating how LLM orchestration can bridge traditionally separate workflows, geophysical inversion, and hydroclimatic data retrieval, which typically require expertise in distinct software ecosystems. This demonstration serves as a proof-of-concept for automated hydroclimatic-geophysical integration. The workflow's ability to autonomously retrieve, synchronize, and correlate climate data with subsurface resistivity changes provides hydrologists with a streamlined approach to understanding how meteorological forcing drives subsur-

face moisture dynamics, a capability that traditionally requires expertise in multiple software ecosystems.

3.3. Data fusion example

We demonstrate multi-method data fusion by integrating seismic refraction with ERT inversion and petrophysical conversion at a granitic hillslope in the Dry Creek Experimental Watershed (Chen & Niu, 2022; Fig. 7). This workflow was initiated by a single comprehensive natural-language request (Fig. 7a) that specified a three-step sequential pipeline: (1) "First, use field seismic refraction data to identify the boundary between regolith and fractured bedrock. Seismic data are in *File_path*," (2) "Then, use this seismic structure to constrain ERT inversion with field ERT data. The ERT data are in *File_path*," and (3) "Finally, convert the resistivity model to water content using layer-specific petrophysical parameters. *Petrophysical properties*." This request demonstrates GAGAW's ability to orchestrate complex, interdependent workflows where the output of one geophysical method (seismic interface) becomes a structural constraint for another (resistivity inversion).

The Context Input Agent parsed the three-step request and generated a workflow configuration that activated the Data Fusion Agent. This agent coordinated multiple specialized sub-agents: the Seismic Processing Agent loaded refraction travel-time data and inverted for velocity structure, then applied a 1000 m/s velocity threshold from the request to extract the regolith-bedrock interface. The Inversion Agent subsequently used this interface as a structural constraint in PyHydro-GeophysX, implementing a structure-constrained inversion where the seismic boundary guides spatial regularization and restricts smoothing across the identified discontinuity. Finally, the Petrophysics Agent applied layer-specific Archie parameters (distinct coefficients above and below the interface) to convert resistivity to water content, with Monte Carlo sampling propagating both petrophysical and structural uncertainty.

Fig. 7c (left panel) shows the inverted seismic velocity model with velocities ranging from ~ 500 m/s to ~ 3000 m/s. The extracted interface (black line at 1000 m/s) delineates a weathering boundary, capturing the asymmetric weathering profile typical of granitic hillslopes where

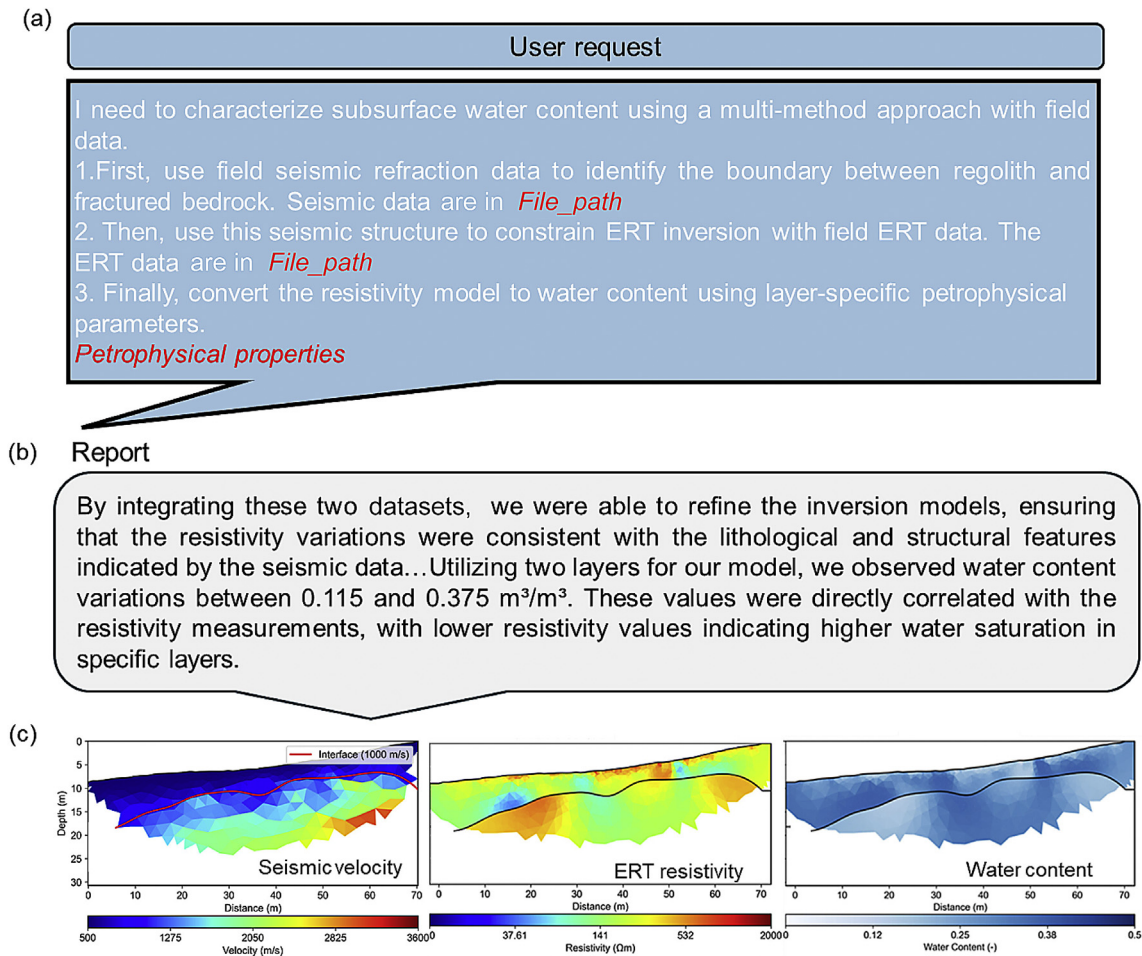


Fig. 7. Structure-constrained data fusion from a single multi-step natural-language request. (a) Comprehensive user prompt specifying a three-stage workflow: seismic interface extraction, structure-constrained ERT inversion, and layer-specific petrophysical conversion. Red italic text indicates user-specified file paths and petrophysical parameters. (b) Excerpt from the automatically generated report. (c) Cross-sectional outputs. Left: Seismic velocity model. Center: Structure-constrained resistivity model. Right: Water content distribution.

aspect, vegetation, and fracture density modulate weathering depth. Velocities below this boundary (>1000 m/s) represent fractured bedrock, while those above (blue, <1000 m/s) correspond to regolith with open porosity and lower mechanical stiffness. Fig. 7c (center panel) presents the resistivity model constrained by the seismic interface. Resistivities span $37.61 \Omega\text{-m}$ to $532 \Omega\text{-m}$, with a sharp transition aligned with the seismic boundary. Critically, the resistivity model exhibits sharper vertical gradients along the interface than would result from standard smooth inversions. Lower resistivities dominate the shallow regolith, consistent with higher porosity and moisture retention, while higher resistivities characterize the deeper fractured bedrock, where reduced porosity limits water storage.

Fig. 7c (right panel) shows the derived water content distribution ranging from 0.12 to 0.38 (volumetric fraction). The Petrophysics Agent applied distinct parameter sets above and below the interface, recognizing that weathered regolith and fractured bedrock require different petrophysical models. Water content values between $0.115 \text{ m}^3/\text{m}^3$ and $0.375 \text{ m}^3/\text{m}^3$ were observed across the two layers (Fig. 7b). The report excerpt (Fig. 7b) notes that “lower resistivity values indicating higher water saturation in specific layers,” correctly interpreting the inverse Archie relationship within each geologically distinct unit.

The automatically generated report (Fig. 7b) synthesizes the fusion workflow, stating: “By integrating these two datasets, we were able to refine the inversion models, ensuring that the resistivity variations are consistent with the lithological and structural features indicated by the seismic data.” This statement illustrates the primary benefit of structure-

constrained inversion: reducing non-uniqueness by incorporating independent geophysical constraints. The seismic interface prevents resistivity smoothing from blurring the weathering boundary, producing a model that is both geophysically consistent (fits ERT data) and geologically plausible (using seismic structure). Monte Carlo uncertainty analysis propagates both petrophysical parameter uncertainty and structural uncertainty from interface picking, though the latter is reduced by the high-contrast velocity boundary. This example illustrates GAGAW’s capacity to orchestrate sequential joint workflows where one geophysical method informs another. The current implementation uses structure-constrained inversion, a pragmatic approach that maintains computational efficiency while incorporating cross-method constraints. This example demonstrates the proof-of-concept for multi-method data fusion, showing that structure-constrained inversion can be automated through natural language prompts while preserving the geophysical rigor of incorporating independent constraints.

4. Discussion

4.1. AQUAH web platform and democratizing geophysical workflows

To lower the barrier to entry beyond command-line interfaces, we are developing Autonomous Query-driven Understanding Agent for Hydrogeophysics (AQUAH, <https://pyhydrogeophysx.streamlit.app/>), a web-based implementation of GAGAW specifically tailored for hydrogeophysical applications and built with Streamlit (Khorasani et al.,

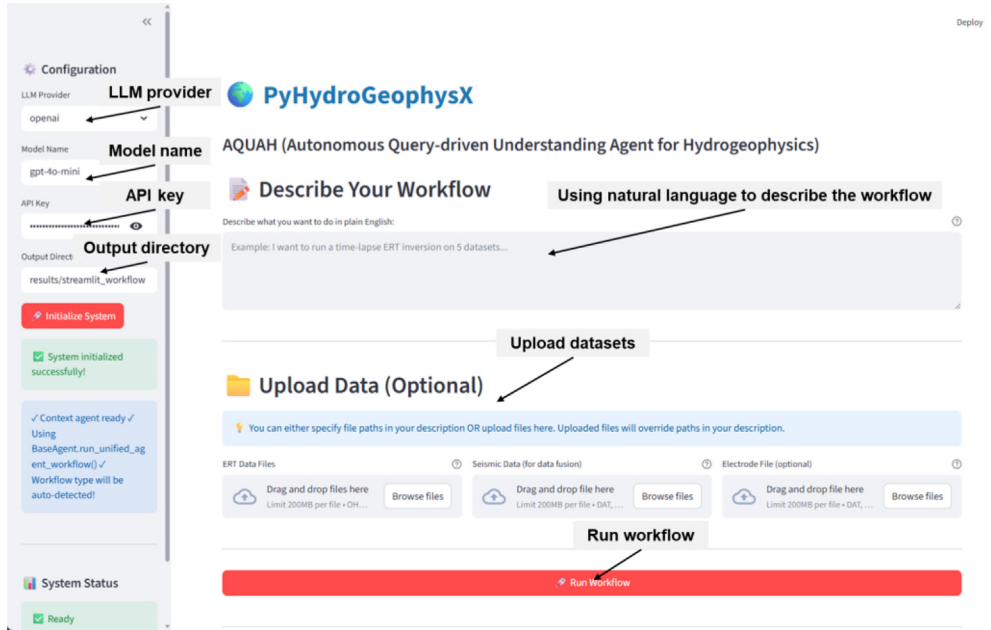


Fig. 8. AQUAH web interface for accessible hydrogeophysical workflow orchestration. The browser-based platform allows users to configure LLM providers (e.g., OpenAI), select models, and input API credentials. Users describe their desired workflow in natural language (e.g., “I want to run a time-lapse ERT inversion to monitor moisture infiltration”) and optionally upload datasets (ERT files, seismic data, and electrode coordinates) through drag-and-drop interfaces. The “Run workflow” button initiates the multi-agent pipeline. Output directories specify where reports and visualizations are saved. This web implementation transforms GAGAW from a command-line tool into an accessible platform for non-programmers, lowering the barrier to sophisticated geophysical analysis for hydrologists, environmental scientists, and educators.

2022) for rapid prototyping and deployment (Fig. 8). The AQUAH interface provides an intuitive graphical environment where users configure their LLM provider and API credentials, specify model preferences (e.g., GPT-4o), and describe their desired workflow in plain English using a guided text box (Fig. 8, “Describe Your Workflow” panel). Users can optionally upload datasets (ERT files, seismic data, and electrode coordinates) directly through drag-and-drop interfaces, with the system automatically detecting file formats and populating configuration fields. Once configured, a single “Run workflow” button initiates the multi-agent pipeline, with real-time status updates (“System initialized successfully!”, “Contact agent ready”, and “Workflow type will be auto-detected”) providing transparency into agent actions. The output directory setting allows users to specify where reports, figures, and intermediate results will be saved.

Streamlit’s reactive framework enables seamless user interaction without requiring web development expertise, allowing the GAGAW team to iterate rapidly on interface design while maintaining a single Python codebase that spans both backend agent logic and frontend presentation. This web interface transforms GAGAW from a tool for Python-fluent researchers into an accessible platform for hydrologists, environmental consultants, and educators who need sophisticated geophysical analysis without programming expertise. Future iterations will add interactive result visualization, workflow templates for common scenarios (e.g., “contamination monitoring”, “groundwater exploration”), and cloud computing integration to handle computationally intensive inversions without local hardware requirements. The Streamlit architecture also facilitates community deployment: users can host AQUAH instances on institutional servers, share workflows via URL parameters, or deploy to cloud platforms like Streamlit Community Cloud for broader accessibility.

4.2. Limitations of the current agent workflow

Across three demonstrations from this work, the agent chain executed end-to-end analyses from plain-language prompts. Several practical issues were mitigated. First, to reduce LLM parsing fragility, a fall-

back extracts key fields when the model returns ill-structured responses (Text S2), preventing run-stopping failures. Second, hierarchical prompt templates and agent-specific schemas improved reproducibility by enforcing consistent defaults such as baseline ordering, regularization, and iteration limits across different agents. Third, heterogeneous data formats and missing metadata were handled by the different agents, for example, integrating external electrode and topography files for the Wyoming case, and climate retrieval used PyDaymet via HyRiver.

However, important limitations still remain. The fusion example uses a sequential, structure-constrained inversion, while other joint or coupled inversion methods are not yet integrated. Geophysics coverage is narrow, centered on ERT, seismic refraction, and basic climate linkage, and does not include other geophysical methods, datasets, and modeling tools, e.g., electromagnetics, borehole data, or integrated hydrological simulators (e.g., Maxwell et al., 2015; Oldenburg et al., 2020; Wang et al., 2025). The current uncertainty quantification propagates only petrophysical parameter uncertainty following the approach of Chen & Niu (2022). Survey errors, including electrode position uncertainty, reciprocal measurement errors, and temperature drift, are not yet incorporated into the final water content uncertainty maps. Integrating these additional error sources into a comprehensive error budget represents an important direction for future development. From an AI standpoint, LLM hallucinations (Zhang et al., 2025) pose risks without stricter schema validation, tests, and version pinning. Error propagation is incomplete: apart from petrophysical Monte Carlo, survey and modeling errors are not consistently carried through to hydrologic targets, and climate–resistivity links are correlational without modeling. Typical failure cases encountered during development include invalid JSON configurations when the LLM returns malformed output and misidentified data formats. These are mitigated by the regex fallback parser, and file parser exceptions. We emphasize that GAGAW is designed to augment, not replace, geophysical expertise. The Report Agent generates summary figures and diagnostic outputs specifically to facilitate expert review, and users should examine inversion convergence, data misfit distributions, and petrophysical parameter selections from the final report before accepting final results. Regarding data privacy, the current

implementation does not transmit raw geophysical data to the LLM API; raw data files remain on the local system and are processed locally by open-source tools. Only metadata, configuration parameters, and natural language descriptions are sent to the API. However, for users with strict data confidentiality requirements, GAGAW supports deployment with local open-source LLMs.

4.3. Future work for agentic geophysics

Building from the limitations above, the near-term roadmap focuses on three complementary directions: technical robustness, methodological expansion, and community accessibility. We will formalize natural-language objectives into typed, multi-agent plans with strict schemas and full traceability, reducing reliance on LLM interpretation flexibility that can introduce variability. Reliability and interpretability will improve through clearly defined workflow standards, comprehensive unit and integration tests, and end-to-end logging that documents every agent decision and parameter choice. We will ground parameter selections with retrieval-augmented generation over technical literature and benchmark datasets, enabling agents to cite methodological precedents when recommending inversion settings or petrophysical ranges. Tighter controls on LLM behavior will come from schema validation and fallback mechanisms that catch hallucinations before they propagate through the workflow. Expanding geophysical coverage will involve incorporating additional open-source platforms enabling workflows that span the full breadth of near-surface geophysics beyond the ERT and seismic focus demonstrated here.

Fig. 1a illustrates how this orchestrator paradigm extends beyond hydrogeophysics. The same agent architecture can coordinate tools for mineral exploration (integrating gravity, magnetics, and induced polarization), geothermal assessment (combining magnetotellurics, seismics, and thermal modeling), or carbon sequestration monitoring (fusing time-lapse seismic and electrical methods with reservoir simulation). The core innovation is not the specific tools being orchestrated, but the orchestration pattern itself: typed configurations, hierarchical agent coordination, uncertainty propagation, and human-in-the-loop review. Since GAGAW leverages existing open-source tools, the computational time for data processing and inversion is unchanged from conventional workflows. The additional time introduced by the agent depends on the LLM API. In this work, we used GPT-4o-mini, which responds quickly as it does not employ extended reasoning. Future implementations using reasoning models with thinking capabilities may have longer response times, but this overhead remains negligible compared to the time required to train new users in geophysical data processing. Learning to process geophysical data from the beginning typically requires weeks of training; with GAGAW, equivalent tasks can be completed within hours by users with no prior geophysical programming experience. Future work could include controlled experiments comparing GAGAW performance with geophysical beginners to quantify the acceleration in learning and productivity.

Critically, this vision does not eliminate the need for expert judgment. Geophysical interpretation requires understanding geological context, recognizing inversion artifacts, and assessing whether model features are data-driven or regularization-imposed. GAGAW and future agent systems will always produce outputs ready for expert review, not autonomous decisions. The paradigm shift is from “experts write code to run tools” to “experts specify goals and validate results”, with agents handling the intermediate orchestration. This preserves physics-based rigor and domain expertise while dramatically expanding who can apply geophysical methods to address pressing societal challenges from groundwater sustainability to climate adaptation to energy transition.

5. Conclusions

This work introduces GAGAW, an agent framework that turns plain-language requests into end-to-end geophysical workflows and validates

them across three hydrogeophysical cases. In the direct ERT water content estimation example, we learn that supplying detailed, site-specific context enables the agent to deliver more accurate and reliable interpretations. The time-lapse analysis example demonstrated that the framework can autonomously integrate climate data with geophysical inversions, revealing strong correlations between precipitation, temperature, and resistivity changes that align with expected hydrological processes. The multi-method data fusion example showed that structure-constrained inversions using seismic interfaces can improve geological consistency and reduce uncertainty in water content estimates.

While current limitations include narrow geophysical coverage, incomplete error propagation, and risks from LLM hallucinations, this work establishes a generalizable agent pattern that fundamentally redefines the role of LLMs in geoscience from passive code assistants to active workflow orchestrators. This paradigm shift can extend beyond hydrogeophysics to resource exploration and solid Earth studies, making sophisticated geophysical analysis accessible to domain experts while preserving essential human oversight and physics-based rigor. By fundamentally improving data accessibility and accelerating geoscientific discovery, GAGAW points toward a future where domain knowledge (not programming skill) determines who can apply geophysical methods for critical decisions.

Declaration of generative AI

During the preparation of this work the author(s) used ChatGPT and Claude in order to grammar checking and coding assistance. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Open research

The datasets used in this work can be found in Uhlemann, Dafflon et al. (2024), Pleasants et al. (2022), and Chen (2025). The report generated by the agent can be found in Chen (2025) and the code is open source at https://github.com/geohang/GAGAW_Hydrogeophysics and <https://github.com/geohang/PyHydroGeophysX>. The AQUAH website is at <https://pyhydrogeophysx.streamlit.app/>. Other open-source codes used in this workflow include pyGIMLi (Rücker et al., 2017) and SimPEG (Cockett et al., 2015), as well as ResIPy (Blanchy et al., 2020).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Hang Chen: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.bdes.2026.100042](https://doi.org/10.1016/j.bdes.2026.100042).

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